

HONEFOSS-HOYBY, Norway

WMO: 014690

Lat: 60.1660N

Lon: 10.2463E

Elev: 459

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	4.7	-3.8	4.6	0.9	0.3	5.7	5.8	15.7	35.8	13.5	35.5	3.3	340	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	81.4	62.5	77.8	61.2	74.0	59.5	64.8	76.4	63.2	73.6	61.5	70.6	5.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	80.8	68.0	59.1	76.1	66.7	57.6	71.9	65.2	30.0	75.6	28.8	73.5	27.6	70.5	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.1	10.5	DB	-1.4	85.8	6.3	3.6	-5.9	88.4	-9.6	90.5	-13.1	92.5	-17.7	95.1	(4)
			WB	-1.9	67.1	6.5	1.8	-6.5	68.4	-10.3	69.4	-13.9	70.4	-18.6	71.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	23.2	25.5	33.4	42.6	51.8	59.1	63.8	60.5	53.4	42.5	33.7	26.3
	DBStd	15.74	9.70	9.33	8.07	6.05	7.17	5.11	5.23	4.10	5.34	6.52	7.30	10.24
	HDD50	3828	832	686	515	233	62	1	0	0	28	247	488	736
	HDD65	8079	1297	1106	979	671	413	190	89	150	347	698	938	1201
	CDD50	1302	0	0	1	12	118	274	428	324	130	14	1	0
	CDD65	78	0	0	0	0	4	13	52	9	0	0	0	0
	CDH74	784	0	0	0	0	72	188	471	53	0	0	0	0
	CDH80	146	0	0	0	0	7	32	106	1	0	0	0	0
Wind	WSAvg	4.9	4.7	4.7	5.2	5.2	5.5	5.2	4.7	4.4	4.7	4.7	5.0	4.5
Precipitation	PrecAvg	30.4	2.1	1.4	1.5	1.8	2.4	3.0	3.4	3.5	3.0	3.2	3.0	1.9
	PrecMax	43.9	4.4	3.4	4.5	3.7	4.9	6.0	6.3	7.0	7.3	8.0	12.3	3.1
	PrecMin	23.4	0.5	0.4	0.0	0.3	0.3	0.4	0.6	1.1	1.2	0.7	0.5	0.4
	PrecStd	4.0	1.0	0.7	0.9	0.9	1.2	1.5	1.2	1.4	1.3	1.7	2.2	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	48.2	59.2	67.2	80.8	84.1	86.4	78.8	72.1	61.5	53.8	49.0
		MCWB	40.8	39.7	46.4	51.6	59.7	62.7	65.0	62.5	59.3	54.4	48.4	43.6
	2%	DB	41.8	43.7	53.8	63.4	75.2	79.7	83.1	75.1	68.3	57.8	50.6	45.9
		MCWB	37.8	37.8	43.2	49.9	57.6	60.1	64.3	61.0	57.1	51.2	45.8	41.8
	5%	DB	38.7	40.5	48.6	59.1	70.5	75.0	79.7	72.0	65.4	55.3	47.5	43.0
		MCWB	35.5	36.3	40.1	47.3	55.0	58.1	63.3	59.9	56.2	50.1	43.8	38.9
	10%	DB	36.1	37.7	44.6	54.7	65.9	70.7	75.6	69.1	62.4	52.8	44.5	39.9
		MCWB	33.4	34.7	37.6	44.3	53.5	56.3	61.2	58.4	54.6	48.2	41.7	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	41.2	47.1	53.5	62.3	63.8	68.0	65.0	61.6	56.1	49.8	44.9
		MCDB	45.0	44.9	58.7	64.1	75.1	78.9	80.8	73.1	67.6	59.4	52.8	48.5
	2%	WB	38.7	38.6	43.7	50.7	59.3	61.7	66.0	63.3	59.2	52.7	46.6	42.1
		MCDB	41.4	42.5	52.6	61.2	71.5	75.0	79.2	70.9	65.0	56.3	49.7	45.4
	5%	WB	35.9	36.7	40.9	48.2	56.8	60.0	64.5	61.7	57.5	50.7	44.2	39.4
		MCDB	38.3	39.9	46.9	56.9	67.9	71.0	76.4	68.7	63.5	54.3	46.9	42.5
	10%	WB	33.6	34.6	38.6	45.6	54.2	58.1	62.8	60.3	55.8	48.7	41.9	36.6
		MCDB	36.1	37.4	43.7	53.8	63.0	67.9	73.3	66.7	61.1	52.3	44.7	39.4
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.7	15.4	18.2	18.6	18.2	17.8	16.2	15.5	11.8	8.8	8.8
		MCDBR	11.1	14.6	21.3	25.6	26.4	25.9	24.4	20.2	19.6	14.0	13.1	12.4
		MCWBR	9.3	10.1	12.9	14.2	12.5	11.3	9.6	9.1	11.0	9.4	10.6	10.4
	5% WB	MCDBR	10.8	12.6	19.2	23.8	22.5	21.4	21.6	16.5	16.2	11.8	12.2	12.0
		MCWBR	9.5	9.2	12.3	13.7	11.2	10.7	9.4	8.5	9.8	9.2	10.5	10.4
Clear-Sky Solar Irradiance	taub	0.245	0.288	0.317	0.360	0.360	0.351	0.371	0.363	0.330	0.315	0.266	0.216	
	taud	2.198	2.285	2.310	2.320	2.420	2.467	2.429	2.445	2.525	2.483	2.292	2.102	
	Ebn at Noon	172	228	258	266	275	279	270	263	255	219	163	133	
	Edh at Noon	12	21	28	33	32	31	32	29	23	17	11	9	
All-Sky Solar Radiation	RadAvg	92	292	762	1219	1532	1737	1551	1220	806	361	121	59	
	RadStd	17	56	76	164	163	140	202	102	97	70	33	5	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	N/A	N/A	+2.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page